

NACS 645 – Reasoning with heuristics and biases

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Modern traditions of rationality

- **Logical rationality** (post-WWII): **maximization of subjective expected utility, consistency axioms, Bayesian probability updating, Nash equilibrium and backward induction.**
 - Researchers sought *pure rationality, valid universally and eternally, independent of the problem at hand, and ideally to be used mechanically by a computer.*
 - Cold War strategists assumed that decision-makers would follow the axioms of logical rationality; original theorists of EU **did not intend these axioms to be universal outside small worlds.**
- **Heuristics-and-biases program** (70s): **deviations from logical rationality were attributed to flaws in human judgment.**
 - Humans are **systematically** and predictably **irrational**.
 - People reduce task complexity (e.g., estimating probabilities) **with heuristics that are economical and efficient** but **systematically violate logical norms**, producing stable patterns of bias.
 - Deviations were taken as evidence that human judgment is error-prone and therefore in need of external steering (e.g., nudging).
 - H-and-B authors treated **logical rationality as a universal norm** for how individuals should decide.
- **Ecological rationality** (90-2000): **rationality is bounded by biological and ecological constraints** such as uncertainty, task complexity and limited cognitive resources.
 - Heuristics function by **purposeful** information restriction.
 - Maximizing utility cannot handle ill-defined situations of **uncertainty** and **intractability**.
 - Biases are not cognitive failures; they appear when *purposeful information restriction* is evaluated under the **wrong normative standard** (i.e., assuming full information and unlimited computation).

Small- and large-worldedness

Table 1. Risk, ambiguity, uncertainty, and intractability

Conditions	Small worlds		Large worlds	
	Risk	Ambiguity	Uncertainty	Intractability
Are all possible future states and consequences of all actions known?	Yes	Yes	No	Yes
Are all probabilities known?	Yes	No	No	Yes
Can optimal action be calculated?	Yes	Yes	No	No

- **Small worlds:** all states, outcomes, and probabilities are known.
Optimization (SEU, Bayesian updating, backward induction) is feasible and normatively appropriate.
- **Large worlds:** states and probabilities are ill-defined, structures are unstable, and computations are intractable.
Optimization breaks down; simple heuristics become the rational strategy.
- Applying small-world norms to large-world decisions produces the appearance of “bias.”

Real-world consequences

- For centuries, irrationality was blamed on passions, ignorance, sloppy thinking, or madness.
- Cold War rationality added 2 obstacles to optimality:
 - **Uncertainty** (unknown unknowns; future states cannot be fully specified)
 - **Intractability** (optimal solutions cannot be computed by humans or machines)
- Under uncertainty and intractability, **optimization and SEU are not applicable**.
- Additionally, Allais and Ellsberg showed that real preferences **violate SEU axioms**.
- Despite that, economists followed the *as-if* doctrine (M. Friedman): people can be modeled **as if** they maximize expected utility.
- This stance led economists to treat SEU as a **predictive ideal**, guiding policy-making for decades.
- A 50-year review concluded that utility-based models have **poor-to-nonexistent** out-of-sample performance (D. Friedman, Isaac & Sunder, 2014).